

Week 7 - Data Ingestion

See the existing documents in MongoDB

The screenshot shows the MongoDB Atlas Search & Vector Search interface. The left sidebar contains navigation options: Search & Vector Search, Search Indexes, AUTO EMBEDDING, Usage, Rate Limits, SHORTCUTS, Cluster Metrics, and Data Explorer. The main content area is titled "Search & Vector Search" and includes a search bar and a "CREATE SEARCH INDEX" button. A yellow banner indicates that the user has reached the 3 search and vector index limit for their M0 free tier cluster, with a link to "Upgrade Cluster". Below this, a table lists the existing search indexes:

Database	Collection	Index Name	Status	Queryable	Type	Index Fields	Documents (Estimated)
db_testleaf	Utestcase	bm25_search	READY	✓	search	"automationManual", "createdBy", "description", "expectedResults", "id", "moduleId", "moduleName", "priority", "risk", "statusCategory", "title", "url", "version"	6,854 (100% indexed of 6,854)
db_testleaf	Utestcase	vector_index_testcaseUI	READY	✓	vectorSearch	"embedding", "id", "moduleId", "moduleName", "description", "expectedResults", "id", "moduleId", "moduleName", "priority", "risk", "statusCategory", "title", "url", "version"	6,854 (100% indexed of 6,854)
db_testleaf	user_stories	vector_index_userstories	READY	✓	vectorSearch	"embedding", "projectName", "statusCategory", "priority", "risk", "title", "url", "version"	190 (100% indexed of 190)

Deleted all existing data's

The screenshot shows the MongoDB Atlas Search & Vector Search interface with a "Get Started" guide. The guide includes the following steps:

- Create Cluster**: Make a cluster to host your data and power your application. [Learn more about cluster creation](#)
- Add Data**: Add data to use in your application. If using Vector Search, your data should include a field containing vector embeddings. The sample_mflixembedded_movies collection includes a field with vector embeddings.
- Create an Index**: Define a search or vector search index to start querying your data. You can define multiple...

The interface also features a "Create Search Index" button and a "Search & Vector Search" section with the text "For AI- and relevance-based search applications" and "Performant and scalable full-text and semantic search as a native database capability." A "Create Search Index" button is also present.

Created the collections called “banking_testcases” and “banking_userstories” and no data available

The image displays two screenshots of the MongoDB Cloud Explorer interface, showing the 'banking_testcases' and 'banking_userstories' collections, both of which are empty.

Top Screenshot: banking_testcases

- URL:** `cloud.mongodb.com/v2/69f210d3a652863c5590d421#/explorer/69f211b22d6f82b11b7ba855/db_testleaf/banking_testcases/...`
- Organization:** Priyanka's Org - 2026...
- Project:** Project 0
- Data Explorer:** Shows the hierarchy: TestLeafPriyanka > db_testleaf > banking_testcases.
- Documents:** 0. The main area displays a message: "This collection has no data. It only takes a few seconds to import data from a JSON or CSV file."
- Buttons:** ADD DATA, UPDATE, DELETE, EXPORT CODE.

Bottom Screenshot: banking_userstories

- URL:** `cloud.mongodb.com/v2/69f210d3a652863c5590d421#/explorer/69f211b22d6f82b11b7ba855/db_testleaf/banking_userstories/...`
- Organization:** Priyanka's Org - 2026...
- Project:** Project 0
- Data Explorer:** Shows the hierarchy: TestLeafPriyanka > db_testleaf > banking_userstories.
- Documents:** 0. The main area displays a message: "This collection has no data. It only takes a few seconds to import data from a JSON or CSV file."
- Buttons:** ADD DATA, UPDATE, DELETE, EXPORT CODE.

Data Ingestion

Created Vector search Index for testcase's

The screenshot shows the MongoDB Atlas Search & Vector Search interface. The left sidebar contains navigation options: Search & Vector Search, Search Indexes, AUTO EMBEDDING, Usage, Rate Limits, SHORTCUTS, Cluster Metrics, and Data Explorer. The main panel displays the 'Search & Vector Search' page with a search bar and a 'CREATE SEARCH INDEX' button. A notification states: 'M0 free clusters have a limit of 3 search and vector indexes per cluster. Indexes Used: 1 of 3.' Below this, a table lists the existing index:

Database	Collection	Index Name	Status	Queryable	Type	Index Fields	Documents (Estimated)
db_testleaf	banking_testcases	vector_index_banking_testcases	READY	✓	vectorSearch	"embedding", "id", "module", "title", "description", "url"	0 (0%) indexed of 0

At the bottom, there are links to 'Learn more about Atlas Search' and 'Learn more about Atlas Vector Search'. The system status is 'All Good'.

Created Vector search index for userstories

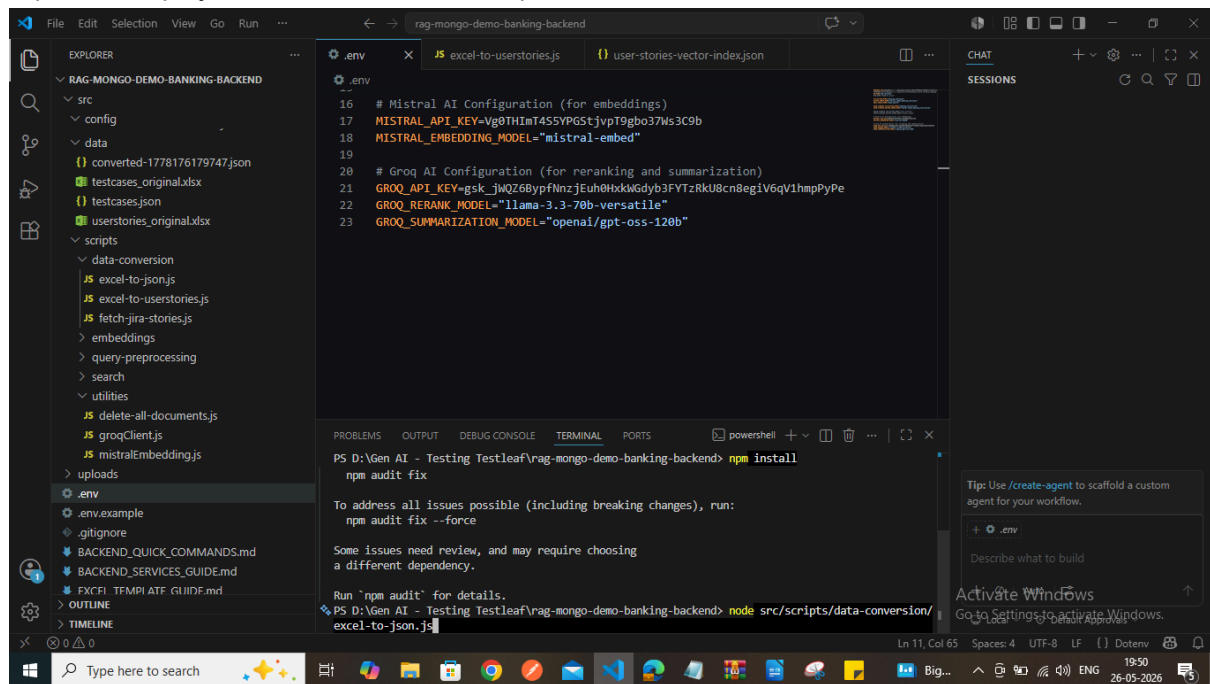
The screenshot shows the MongoDB Atlas Search & Vector Search interface with two indexes listed. The notification now states: 'M0 free clusters have a limit of 3 search and vector indexes per cluster. Indexes Used: 2 of 3.' The table lists the following indexes:

Database	Collection	Index Name	Status	Queryable	Type	Index Fields	Documents (Estimated)
db_testleaf	banking_testcases	vector_index_banking_testcases	READY	✓	vectorSearch	"embedding", "id", "module", "title", "description", "url"	0 (0%) indexed of 0
db_testleaf	banking_userstories	vector_index_banking_userstories	READY	✓	vectorSearch	"embedding", "jiraMetadata.projectKey", "status.name", "priority.name", "assignee.displayName", "url"	0 (0%) indexed of 0

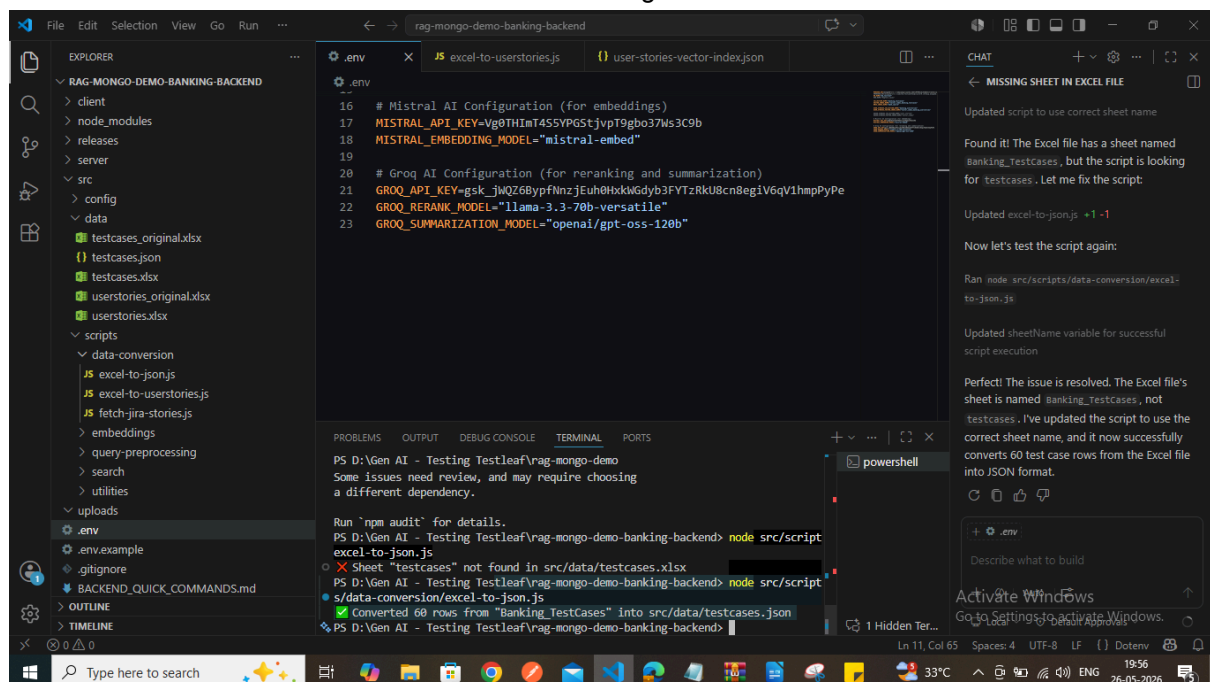
The system status remains 'All Good'.

Data Ingestion for banking testcases had been done from backend

Imported the project in VS code and done npm install



Added comment to convert data and create embeddings for testcases via backend



Added comment to create embeddings and store data for testcases

VS Code interface showing the initial state of the project. The Explorer panel on the left shows the file structure. The main editor shows the 'excel-to-userstories.js' file with a comment about creating embeddings. The terminal shows the execution of the script, which successfully processes 60 test cases. The Chat panel on the right shows a conversation about a missing sheet in the Excel file.

```
PS D:\Gen AI - Testing Testleaf\rag-mongo-demo-banking-backend> node src/scripts/data-conversion/excel-to-json.js
PS D:\Gen AI - Testing Testleaf\rag-mongo-demo-banking-backend> node src/scripts/s/embeddings/create-embeddings-batch-mistral.js
Input files: src/data/testcases.json
MISTRAL AI BATCH PROCESSING: 60 test cases
Using Mistral AI Embedding API for Cost-Effective Processing!
Configuration:
  Batch Size: 50 testcases per API call
  Concurrent Batch Calls: 3
  MongoDB Batch Size: 100
  Delay Between Batch Groups: 1000ms
  API: Mistral AI (mistral-embed)
  Embedding Dimensions: 1024
  Database: db_testleaf
  Collection: banking_testcases
  Total Batches: 2
  Batch Groups: 1
  Estimated Time: 0.1 minutes
[Batch 1/2] Processing 50 testcases with Mistral AI...
[Batch 2/2] Processing 10 testcases with Mistral AI...
[Batch 2/2] Success! Cost: $0.000096 | Tokens: 964
[Batch 1/2] Success! Cost: $0.000481 | Tokens: 4811
Progress: 60/60 (100.0%) | Rate: 17.6/sec | ETA: 0s | Cost: $0.000578
MISTRAL AI BATCH PROCESSING COMPLETE!
Final Statistics:
  Total Time: 4s
  Processing Rate: 13.6 testcases/second
  Total Test Cases: 60
```

VS Code interface showing the final state of the project. The Explorer panel on the left shows the file structure. The main editor shows the 'excel-to-userstories.js' file with a comment about creating embeddings. The terminal shows the execution of the script, which successfully processes 60 test cases. The Chat panel on the right shows a conversation about a missing sheet in the Excel file.

```
PS D:\Gen AI - Testing Testleaf\rag-mongo-demo-banking-backend> node src/scripts/s/embeddings/create-embeddings-batch-mistral.js
[Batch 2/2] Processing 10 testcases with Mistral AI...
[Batch 2/2] Success! Cost: $0.000096 | Tokens: 964
[Batch 1/2] Success! Cost: $0.000481 | Tokens: 4811
Progress: 60/60 (100.0%) | Rate: 17.6/sec | ETA: 0s | Cost: $0.000578
MISTRAL AI BATCH PROCESSING COMPLETE!
Final Statistics:
  Total Time: 4s
  Processing Rate: 13.6 testcases/second
  Total Test Cases: 60
  Successfully Processed: 60
  Failed: 0
  Success Rate: 100.0%
  Total Cost: $0.000578
  Total Tokens: 5,775
  Average Cost per Test Case: $0.00000963
  Average Tokens per Test Case: 96
  Cost Savings: ~90% cheaper than OpenAI embeddings
PS D:\Gen AI - Testing Testleaf\rag-mongo-demo-banking-backend>
```

Verified the data stored in mongoDB

The screenshot displays the MongoDB Atlas Search & Vector Search interface. The left sidebar shows the 'Search & Vector Search' section with options for 'AUTO EMBEDDING', 'Usage', 'Rate Limits', 'SHORTCUTS', 'Cluster Metrics', and 'Data Explorer'. The main area is titled 'Vector Search' and shows a table of indexes.

Collection	Index Name	Status	Queryable	Type	Index Fields	Documents (Estimated)	Size	Required Memory
banking_testcases	vector_index_banking_testcases	READY	✓	vectorSearch	"embedding": "id", "module": "title", "description": "2,20208"	60 (100%) indexed of 60	3228	3228
banking_userstories	vector_index_banking_userstories	READY	✓	vectorSearch	"embedding": "jiraMetadata.projectKey", "status.name", "priority.name", "assignee.displayName", "2,20208"	0 (0%) indexed of 0	3248	3248

Buttons for 'VIEW COMPOUND QUERY EXAMPLE' and '+ CREATE SEARCH INDEX' are visible. A blue banner at the bottom states: 'Users have a limit of 3 search and vector indexes per cluster. Indexes Used: 2 of 3.'

Health Check in Postman

The screenshot shows the Postman interface with a workspace named 'My Workspace'. The left sidebar lists collections, including 'RAG MongoDB - Data Ingestion' which contains a 'GET 1. Health Check' request. The main area shows the details of this request, which is a GET request to '[[baseUri]] /api/health'. The response is a 200 OK status with a response time of 493.20 ms. A 'Response Time' breakdown is shown:

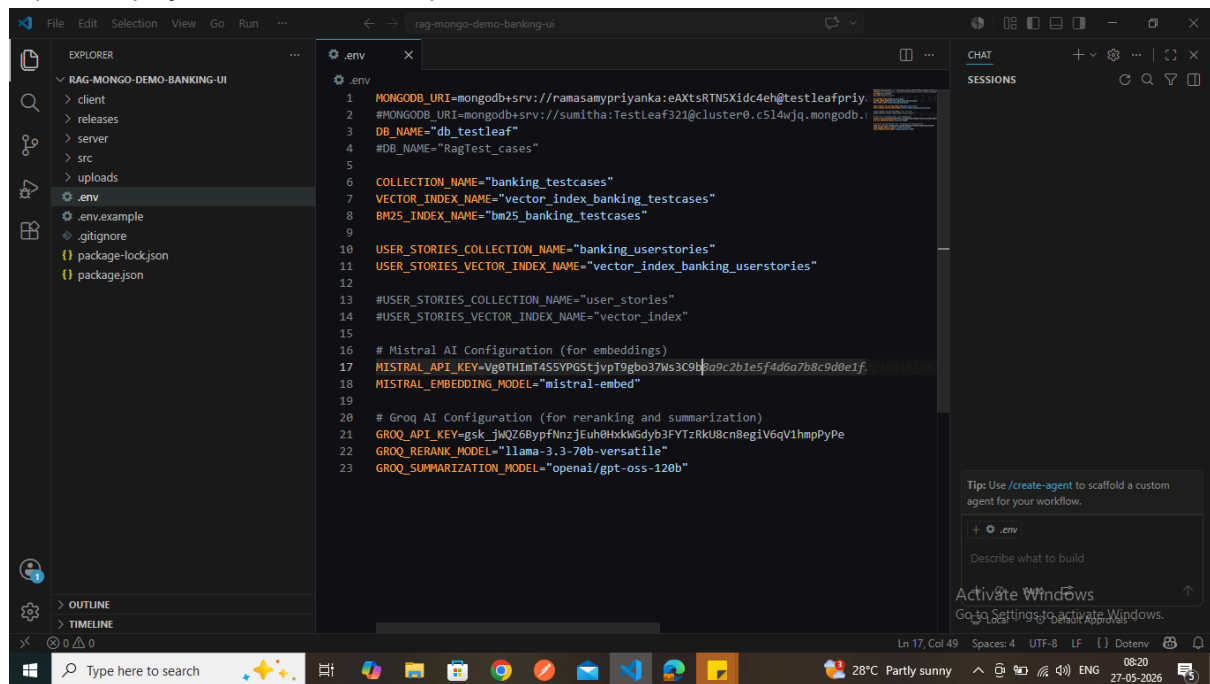
Step	Time (ms)
Prepare	48.33
Socket Initialization	26.91
DNS Lookup	2.98
TCP Handshake	1
Waiting (TTFB)	446.20
Download	15.68
Process	1.91

The response body is shown as JSON:

```
{ "status": "success", "message": "Health check passed" }
```

Data ingestion had been done from FrontEnd UI for banking testcases and userstories

Import the project in VS Code and updated the .env file

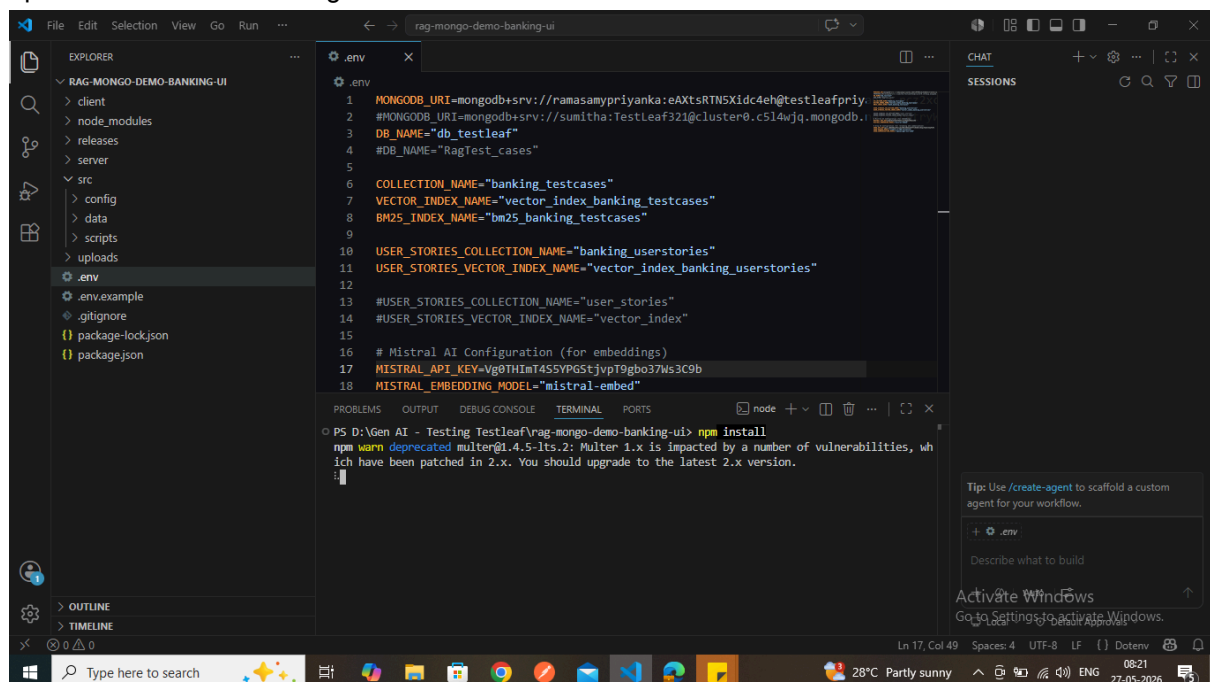


The screenshot shows the Visual Studio Code interface with the `.env` file open. The file contains the following configuration:

```
1 MONGODB_URI=mongodb+srv://ramasamypryanka:eAXtsRTN5Xidc4eh@testleafpriy
2 #MONGODB_URI=mongodb+srv://sumitha:TestLeaf321@cluster0.c514wj.mongodb.
3 DB_NAME="db_testleaf"
4 #DB_NAME="RagTest_cases"
5
6 COLLECTION_NAME="banking_testcases"
7 VECTOR_INDEX_NAME="vector_index_banking_testcases"
8 BM25_INDEX_NAME="bm25_banking_testcases"
9
10 USER_STORIES_COLLECTION_NAME="banking_userstories"
11 USER_STORIES_VECTOR_INDEX_NAME="vector_index_banking_userstories"
12
13 #USER_STORIES_COLLECTION_NAME="user_stories"
14 #USER_STORIES_VECTOR_INDEX_NAME="vector_index"
15
16 # Mistral AI Configuration (for embeddings)
17 MISTRAL_API_KEY=Vg0THimT455YPGStjvpT9gbo37Ws3C9bBa9c2b1e5f4d6a7b8c9d0e1f
18 MISTRAL_EMBEDDING_MODEL="mistral-embed"
19
20 # Groq AI Configuration (for reranking and summarization)
21 GROQ_API_KEY=gsk_jWQZ6BypfNnzjEuh0HxkWgdyb3FYtZrKtU8cn8EgiV6qV1hmpPyPe
22 GROQ_RERANK_MODEL="llama-3.3-70b-versatile"
23 GROQ_SUMMARIZATION_MODEL="openai/gpt-oss-120b"
```

The Explorer sidebar on the left shows the project structure for `RAG-MONGO-DEMO-BANKING-UI`, including `client`, `releases`, `server`, `src`, `uploads`, `.env`, `.env.example`, `.gitignore`, `package-lock.json`, and `package.json`. The bottom status bar indicates the file is at line 17, column 49, with 4 spaces, UTF-8 encoding, LF line endings, and the Dotenv extension is active.

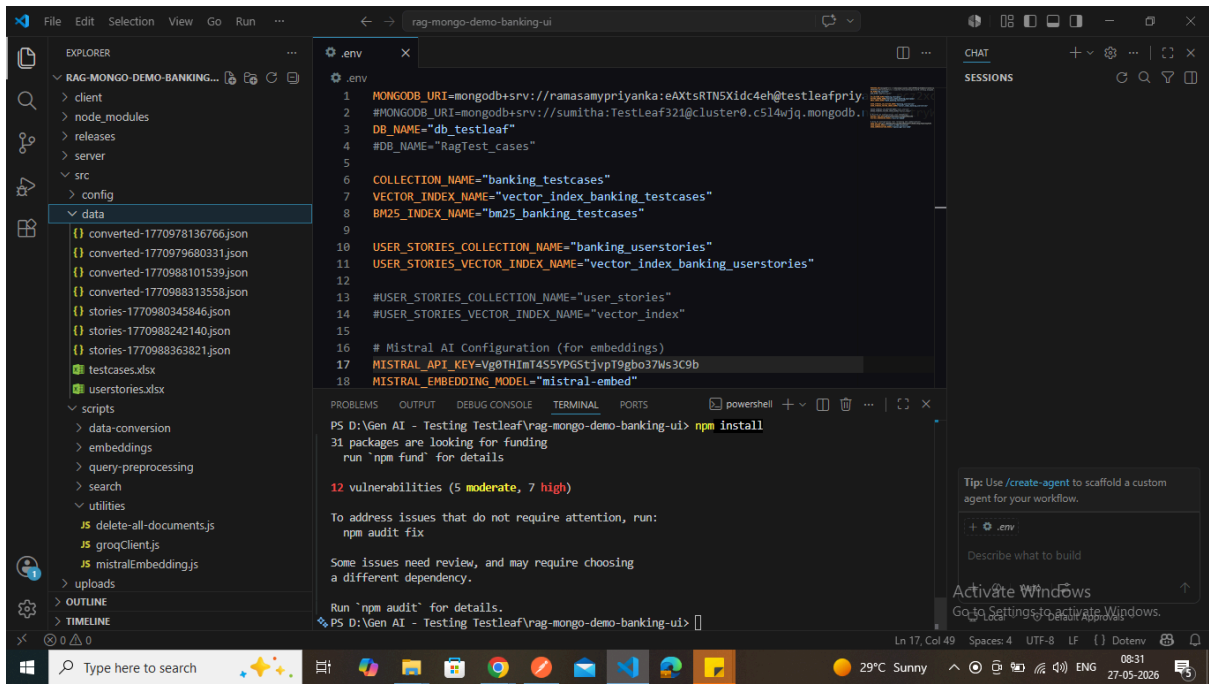
Npm install command were given in Terminal



The screenshot shows the Visual Studio Code interface with the `.env` file open and the Terminal panel at the bottom. The terminal shows the command `npm install` being executed in the `rag-mongo-demo-banking-ui` directory. The output is:

```
PS D:\Gen AI - Testing Testleaf\rag-mongo-demo-banking-ui> npm install
npm warn deprecated multier@1.4.5-lts.2: Multier 1.x is impacted by a number of vulnerabilities, wh
ich have been patched in 2.x. You should upgrade to the latest 2.x version.
```

The Explorer sidebar on the left shows the project structure, with `node_modules` now visible under the `client` directory. The bottom status bar indicates the file is at line 17, column 49, with 4 spaces, UTF-8 encoding, LF line endings, and the Dotenv extension is active.



Existing data in mongoDB

The screenshot shows the MongoDB Atlas Search & Vector Search interface. The table below lists the existing search indexes:

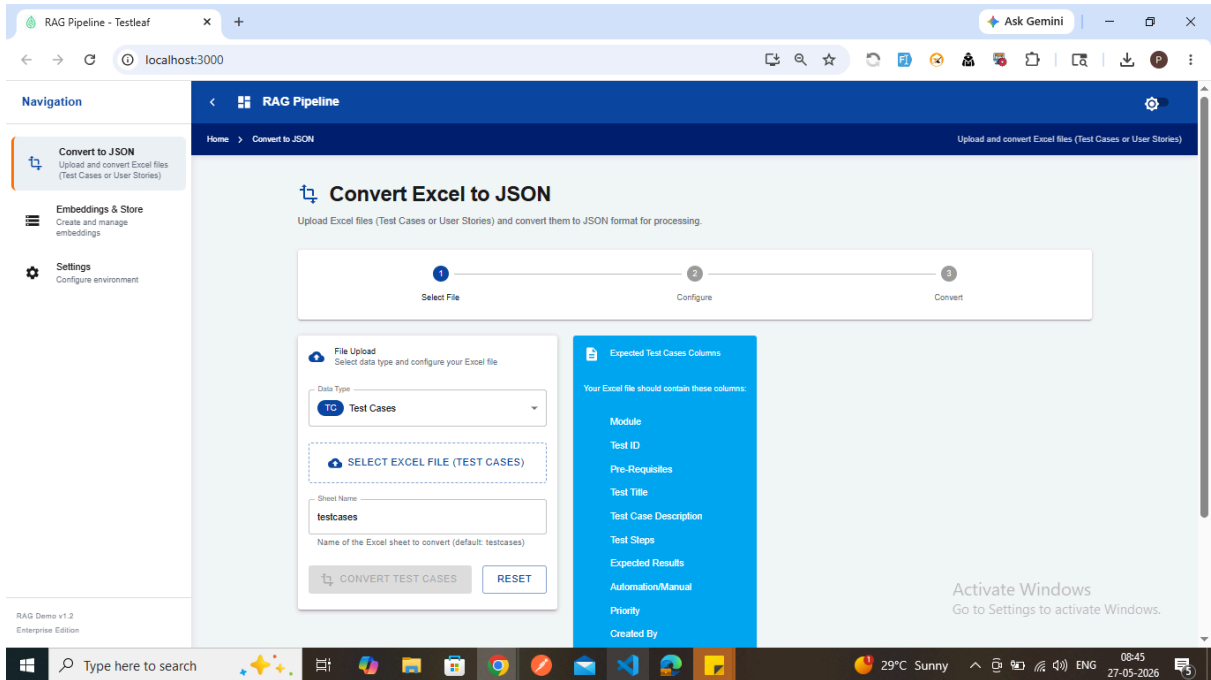
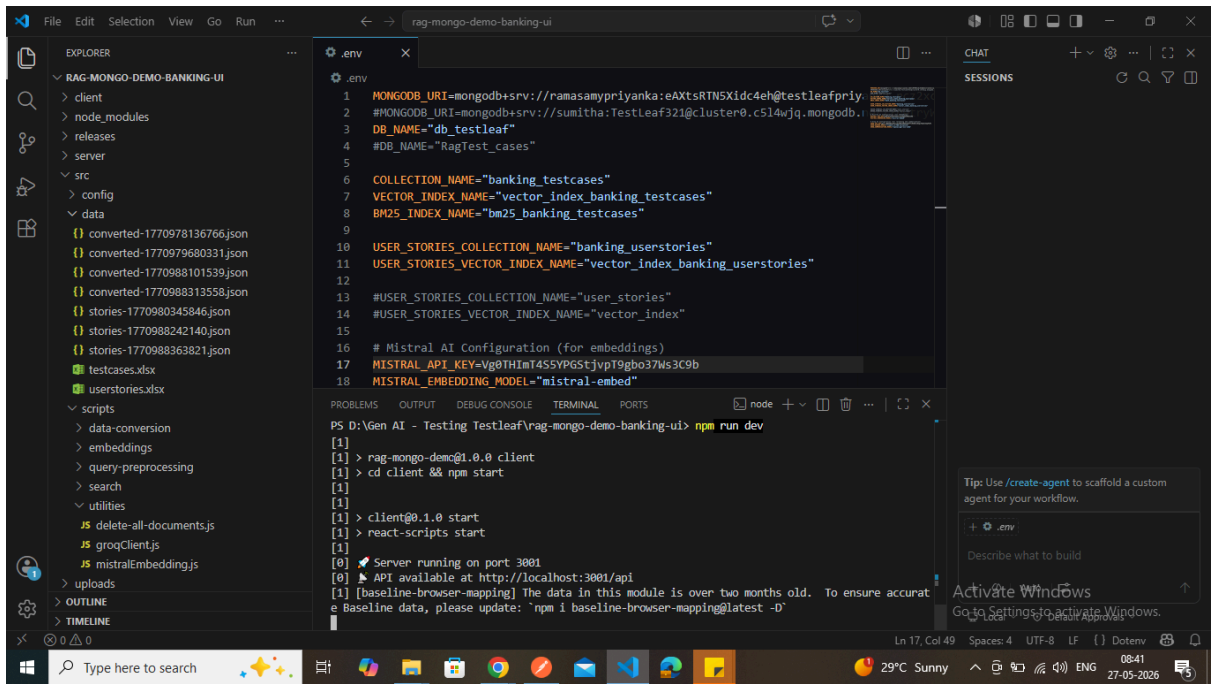
Database	Collection	Index Name	Status	Queryable	Type	Index Fields	Documents (Estimated)	Size
db_testleaf	banking_testcases	bm25_banking_testcases	READY	✓	search	"automationManual", "createdBy", "description", "expectedResults", "id", "p rimaryKey"	180 (100%) indexed of 180	0B
db_testleaf	banking_testcases	vector_index_banking_testcases	READY	✓	vectorSearch	"embedding", "id", "module", "title", "description", "p rimaryKey"	180 (100%) indexed of 180	516.97K
db_testleaf	banking_userstories	vector_index_banking_userstories	READY	✓	vectorSearch	"embedding", "jiraMetadata.projectKey", "status.name", "priority.name", "assignee.displayName", "p rimaryKey"	9 (100%) indexed of 9	55.7KB

The image shows a Windows 11 desktop with a dark-themed interface. The primary application is Visual Studio Code, which has a file explorer on the left showing a project named 'RAG-MONGO-DEMO-BANKING-UI'. The main editor area displays a file named '.env' containing a MongoDB connection string and various configuration variables like 'DB_NAME', 'COLLECTION_NAME', and 'MISTRAL_API_KEY'. A terminal window is open at the bottom, showing the command 'npm install' and its output, which includes a warning about vulnerabilities and instructions to run 'npm audit fix'. The taskbar at the bottom features several pinned applications, including a search bar, File Explorer, and various web browsers. The system tray on the right shows the date and time as '08:36 27-05-2026'.

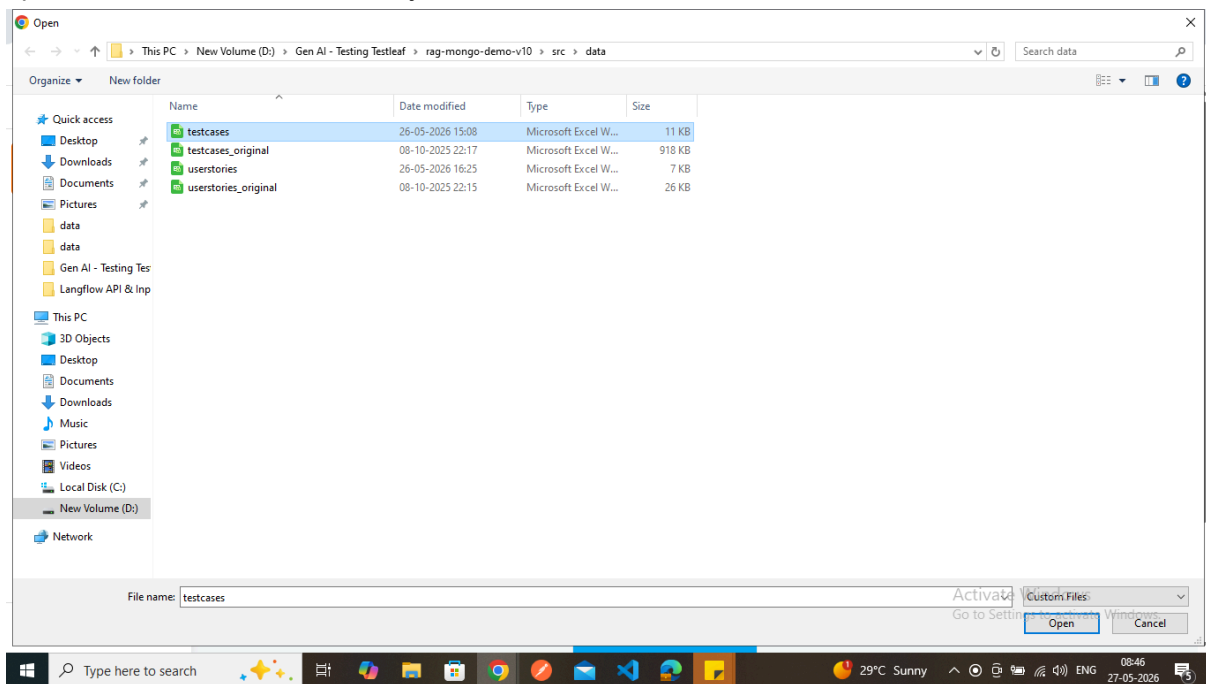
The screenshot shows the MongoDB Atlas Search interface. On the left is a sidebar with navigation options: Search & Vector Search, Search Indexes, AUTO EMBEDDING, Usage, Rate Limits, SHORTCUTS, Cluster Metrics, and Data Explorer. The main panel displays a table of search indexes. A yellow banner at the top indicates a free tier limit. The table has columns for Index Name, Status, Queryable, Type, Index Fields, Documents (Estimated), Size, Required Memory, and Actions. Three indexes are listed, all with a 'READY' status. The first index, '25_banking_testcoses', is of type 'search'. The second, 'stor_index_banking_testcoses', and the third, 'stor_index_banking_userstories', are of type 'vectorSearch'.

Index Name	Status	Queryable	Type	Index Fields	Documents (Estimated)	Size	Required Memory	Actions
25_banking_testcoses	READY View status details	✓	search	"automationManual", "createdBy", "description", "expectedResults", "id", "isMore"	0 (0%) Indexed of 0	323B	-	QUERY ...
stor_index_banking_testcoses	READY View status details	✓	vectorSearch	"embedding", "id", "module", "title", "description", "isMore"	0 (0%) Indexed of 0	323B	323B	QUERY ...
stor_index_banking_userstories	READY View status details	✓	vectorSearch	"embedding", "firstMetadata.projectKey", "status.name", "priority.name", "assignee.displayName", "isMore"	9 (100%) Indexed of 9	55.7KB	55.7KB	QUERY ...

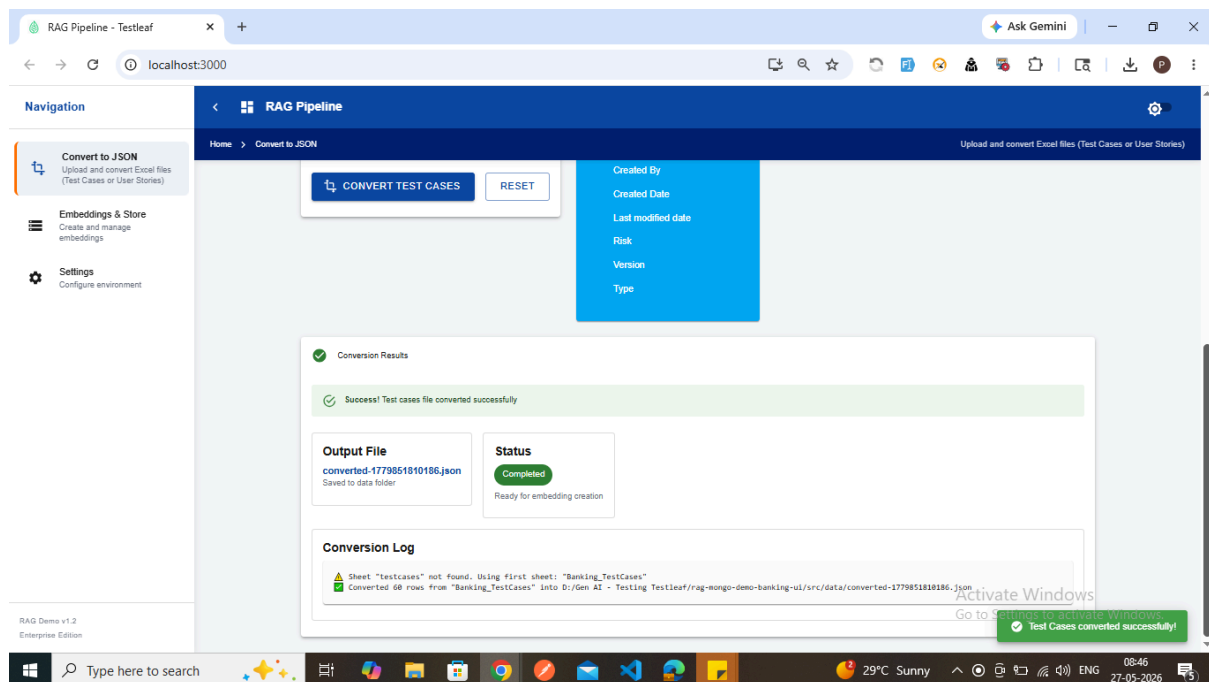
Learn more about Atlas Search
Learn more about Atlas Vector Search



Upload the testcases in convert to json one



Converted the testcases



Go to embeddings and selected the converted json file

The screenshot shows the 'RAG Pipeline - Testleaf' application interface. The left sidebar contains navigation options: 'Convert to JSON', 'Embeddings & Store', and 'Settings'. The main content area is titled 'Embeddings & Store' and displays a table of 'Available Files'. The table has columns for File Name, Size, Modified, and Type. The file 'converted-1779851810186.json' is selected. The 'Actions' panel on the right shows 'PROCESSING...' and 'Debug Info'.

File Name	Size	Modified	Type
converted-1779851810186.json	44.38 KB	5/27/2028 8:48:51 AM	JSON
stories-1770988383821.json	198.29 KB	2/13/2028 8:42:45 PM	JSON
converted-1770988313558.json	8.28 MB	2/13/2028 8:41:57 PM	JSON
stories-1770988242140.json	198.29 KB	2/13/2028 8:40:44 PM	JSON

Selected 1 file(s): converted-1779851810186.json

Debug Info:
Selected Files: 1 (converted-1779851810186.json)
Processing: true
Button Disabled: true
Files loaded: 8

Processing Time: ~10.30 seconds per file
Cost: ~\$0.0001 per test case

Embedding is done

The screenshot shows the 'RAG Pipeline - Testleaf' application interface. The left sidebar contains navigation options: 'Convert to JSON', 'Embeddings & Store', and 'Settings'. The main content area is titled 'Embeddings & Store' and displays a table of 'Available Files'. The file 'converted-1779851810186.json' is selected. The 'Actions' panel on the right shows 'PROCESSING...' and 'Debug Info'.

File Name	Size	Modified	Type
converted-1779851810186.json	44.38 KB	5/27/2028 8:48:51 AM	JSON
stories-1770988383821.json	198.29 KB	2/13/2028 8:42:45 PM	JSON
converted-1770988313558.json	8.28 MB	2/13/2028 8:41:57 PM	JSON
stories-1770988242140.json	198.29 KB	2/13/2028 8:40:44 PM	JSON

Selected 1 file(s): converted-1779851810186.json

Processing Results
1 completed, 0 failed

testcases completed

Input file: src/data/converted-1779851810186.json
MISTRAL AI BATCH PROCESSING: 50 test cases
Using Mistral AI Embedding API for Cost-Effective Processing!
Configuration:
Batch Size: ...

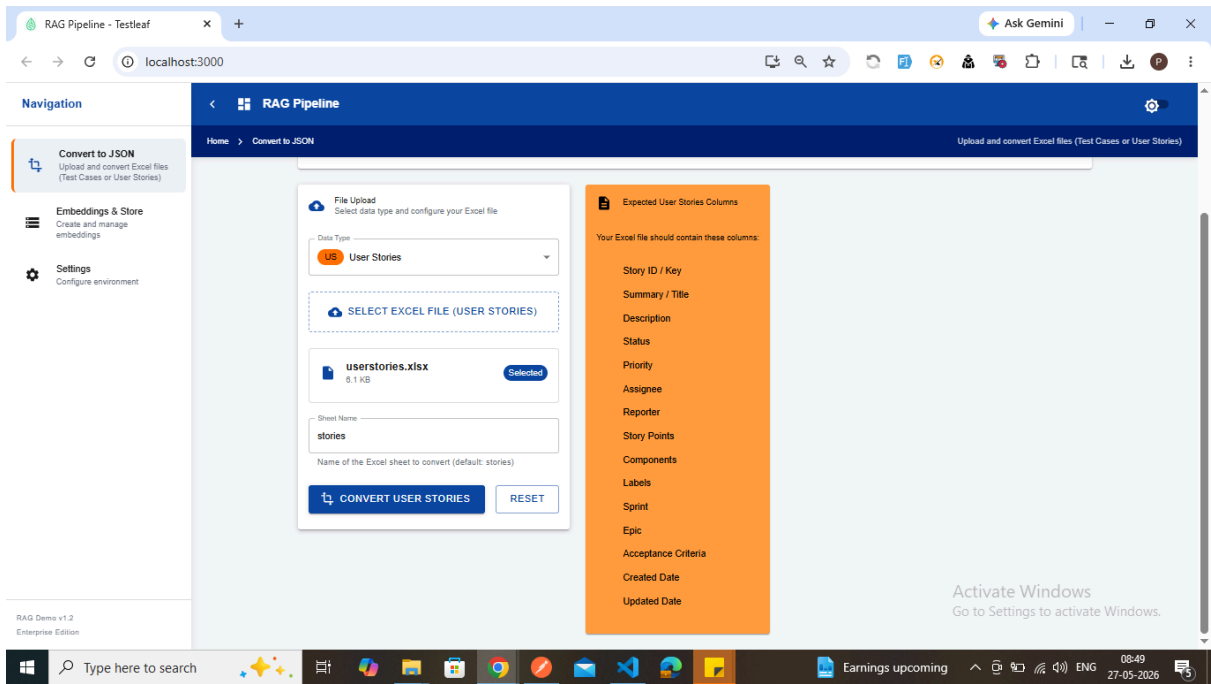
Testcase's had been stored in mongoDB

The screenshot displays the MongoDB Atlas Search & Vector Search interface. The left sidebar shows navigation options like 'Search Indexes', 'AUTO EMBEDDING', 'Usage', 'Rate Limits', 'SHORTCUTS', 'Cluster Metrics', and 'Data Explorer'. The main panel shows a table of search indexes for the 'db_testleaf' database. A yellow banner at the top indicates that the 3 search and vector index limit for the M0 free tier cluster has been reached, with a link to 'Upgrade Cluster'.

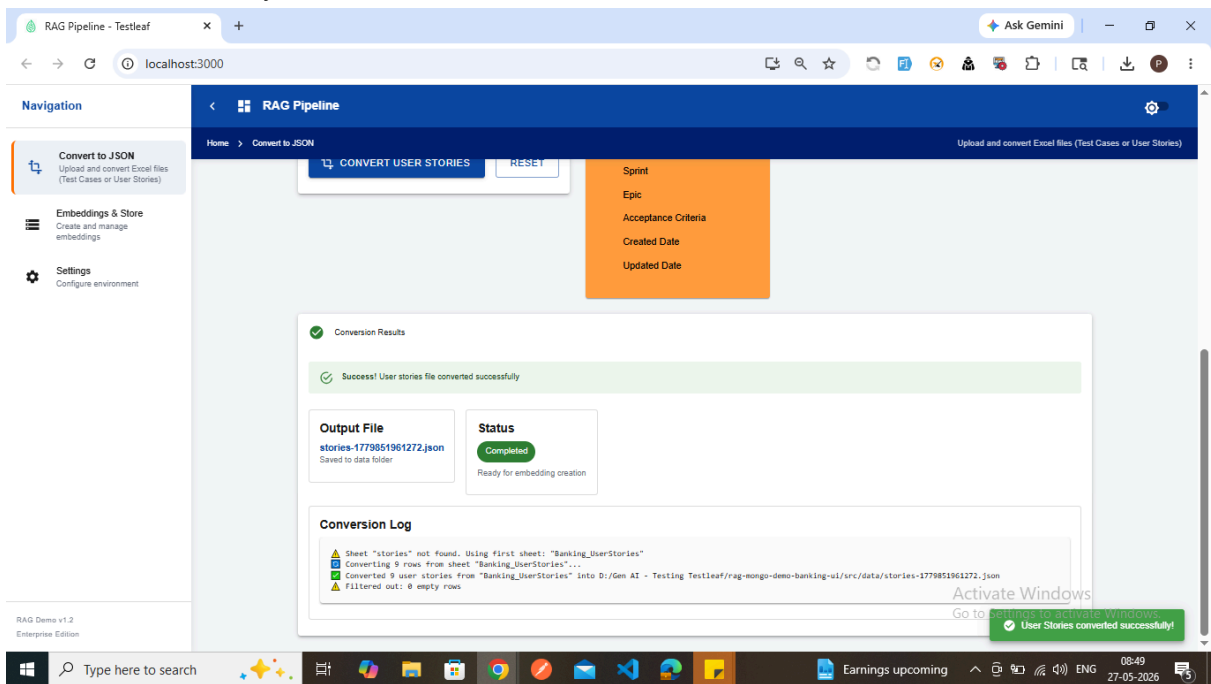
Database	Collection	Index Name	Status	Queryable	Type	Index Fields	Documents (Estimated)	Size
db_testleaf	banking_testcases	bm25_banking_testcases	READY	✓	search	"automationManual", "createdBy", "description", "expectedResults", "id", "priorityName", "statusName", "title", "updatedAt"	60 (100%) indexed of 60	323B
db_testleaf	banking_testcases	vector_index_banking_testcases	READY	✓	vectorSearch	"embedding", "id", "module", "title", "description", "priorityName", "statusName"	60 (100%) indexed of 60	323B
db_testleaf	banking_userstories	vector_index_banking_userstories	READY	✓	vectorSearch	"embedding", "jiraMetadata.projectKey", "statusName", "priorityName", "assignee.displayName", "description"	9 (100%) indexed of 9	55.7KB

Similarly going to upload userstory in UI

The screenshot shows a Windows File Explorer window with the path 'This PC > New Volume (D:) > Gen AI - Testing Testleaf > rag-mongo-demo-v10 > src > data'. The file list shows four files: 'testcases' (11 KB), 'testcases_original' (918 KB), 'userstories' (7 KB), and 'userstories_original' (26 KB). The 'userstories' file is selected. The bottom of the window shows the 'File name' field with 'userstories' and buttons for 'Open' and 'Cancel'.



Converted successfully



Going to do embedding and selected the converted stories.json file and clicked create embeddings

The screenshot shows the 'RAG Pipeline - Testleaf' application interface. The 'Navigation' sidebar on the left includes 'Convert to JSON', 'Embeddings & Store', and 'Settings'. The main content area is titled 'Embeddings & Store' and displays a table of 'Available Files' (9 JSON files found). The file 'stories-1779851961272.json' is selected. The 'Actions' panel on the right shows the 'CREATE EMBEDDINGS' button and a 'Debug Info' section with details about the selected file and processing status.

File Name	Size	Modified	Type
stories-1779851961272.json	8.98 KB	5/27/2028 8:49:22 AM	JSON
converted-1779851810188.js	44.38 KB	5/27/2028 8:48:51 AM	JSON
stories-1770988383821.json	188.29 KB	2/13/2028 8:42:45 PM	JSON
converted-1770988313558.js	8.28 MB	2/13/2028 8:41:57 PM	JSON

Selected 1 file(s): stories-1779851961272.json

Embedding completed

The screenshot shows the 'RAG Pipeline - Testleaf' application interface after the embedding process is complete. The 'Processing Results' section displays a green checkmark and the message 'user-stories completed'. The 'Embeddings & Store' section shows the file 'stories-1779851961272.json' in the 'Selected Files' list. A green banner at the bottom right indicates 'Successfully processed all 1 files!'.

Processing Results
1 completed, 0 failed

user-stories completed

Input file: src/data/stories-1779851961272.json
✓ MISTRAL AI BATCH PROCESSING: 9 user stories
✓ Using Mistral AI Embedding API for Cost-Effective Processing!
Configuration:
Batch Size: 5...

Selected Files
stories-1779851961272.json

Successfully processed all 1 files!

Testleaf

(5) WhatsApp

TestLeafPriyanka

Week 8 - Retrieval

Week 7 - Ingestion

New Tab

Ask Gemini

cloud.mongodb.com/v2/69f210d3a652863c5590d421#/clusters/atlasSearch/TestLeafPriyanka

Search & Vector Search

Find a search index

VIEW COMPOUND QUERY EXAMPLE

CREATE SEARCH INDEX

You have reached the 3 search and vector index limit for your M0 free tier cluster. Upgrade Cluster

Database	Collection	Index Name	Status	Queryable	Type	Index Fields	Documents (Estimated)	Size
db_testleaf	banking_testcooses	bm25_banking_testcooses	READY	✓	search	"automationManual", "createdBy", "description", "expectedResults", "id", "title"	60 (100%) indexed of 60	323B
db_testleaf	banking_testcooses	vector_index_banking_testcooses	READY	✓	vectorSearch	"embedding", "id", "module", "title", "description"	60 (100%) indexed of 60	323B
db_testleaf	banking_userstories	vector_index_banking_userstories	READY	✓	vectorSearch	"embedding", "jiraMetadata.projectKey", "status.name", "priority.name", "assignee.displayName"	9 (50%) indexed of 18	55.7KB

Learn more about Atlas Search

Learn more about Atlas Vector Search

System Status: All Good

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Activate Windows

Go to Settings to activate Windows.

Type here to search

Humid

08:51

27-05-2026